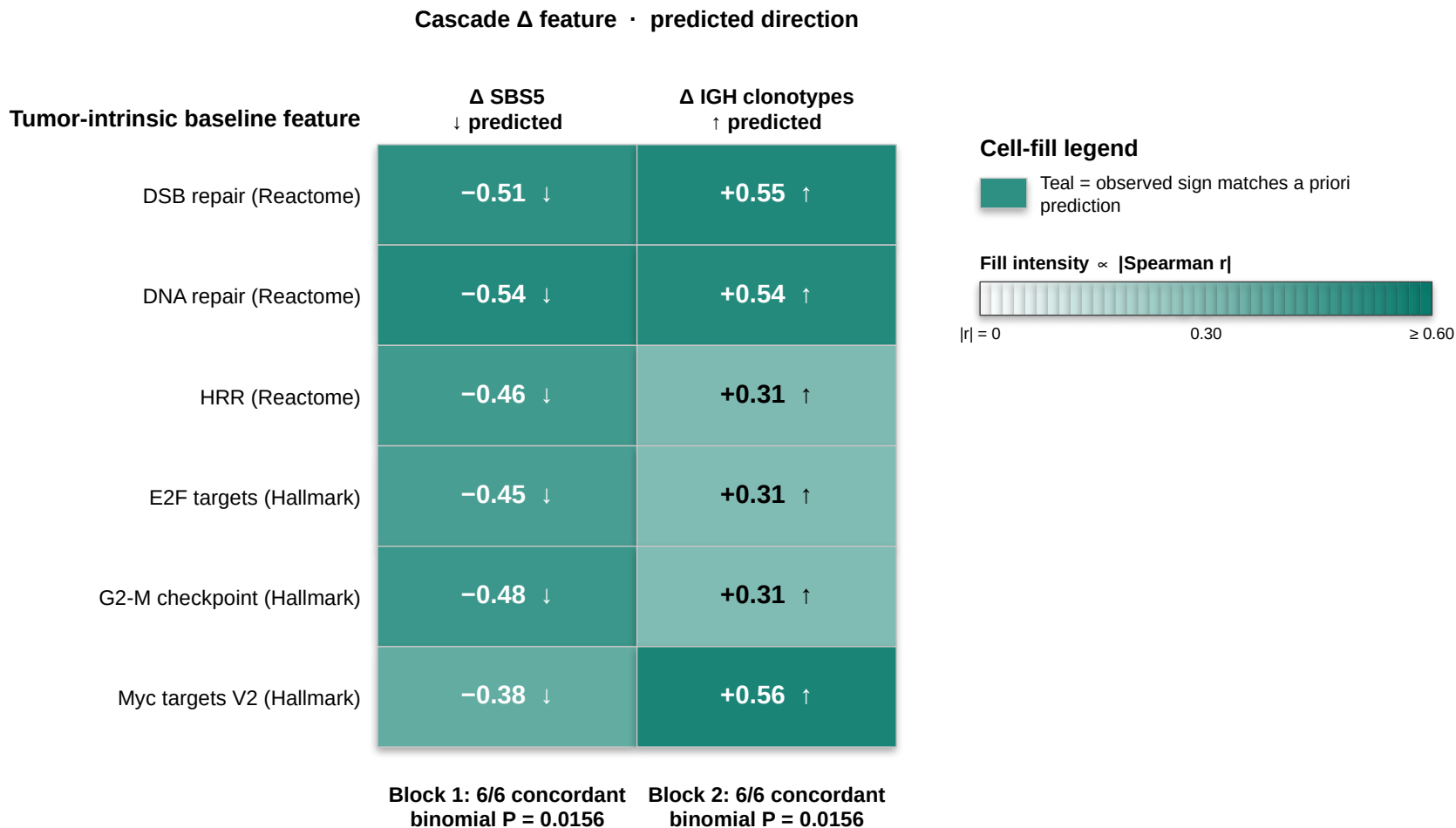


A

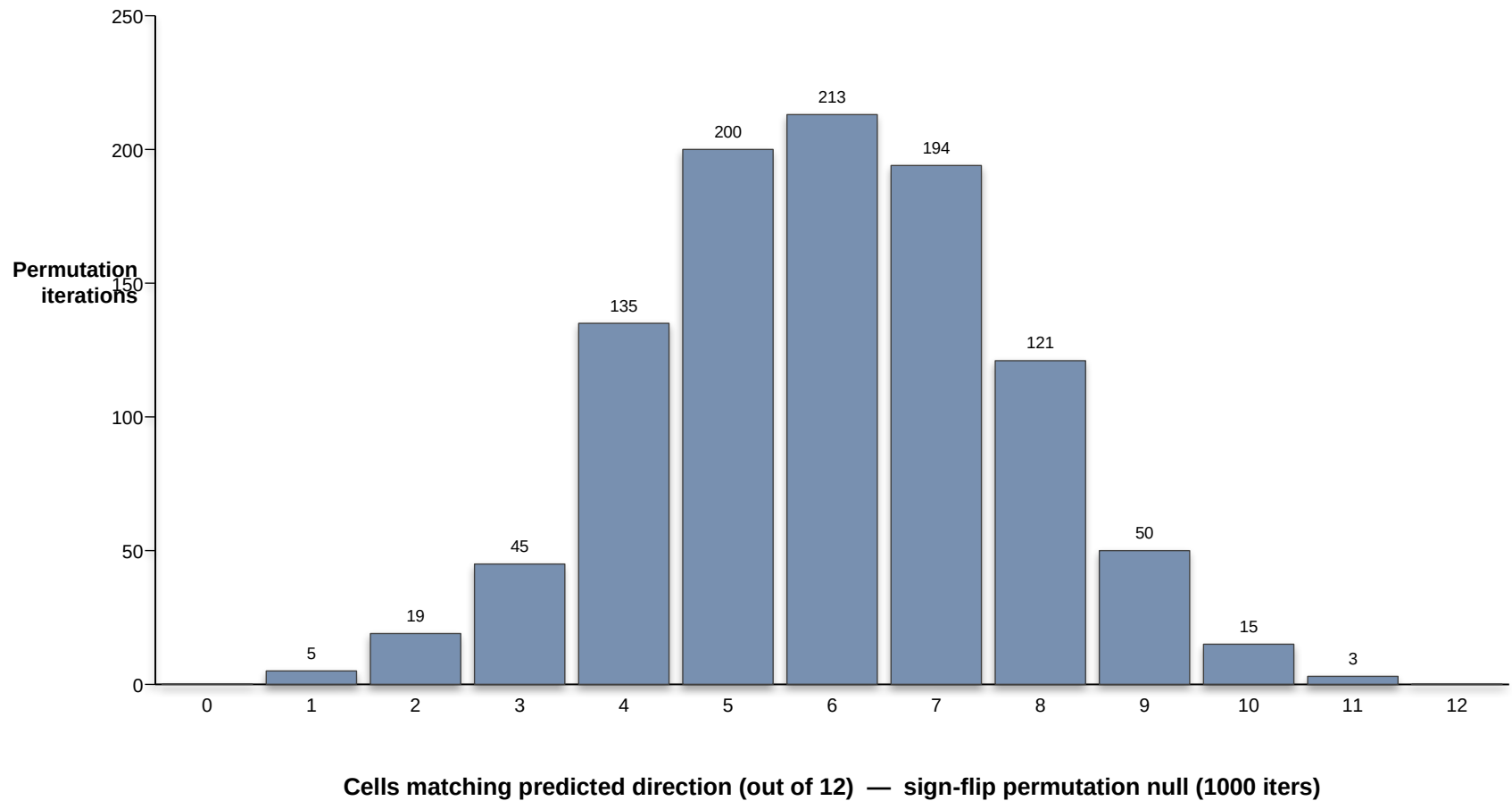


Combined block: 12/12 cells in predicted direction. One-sided binomial $P = 0.000244$ ($= 1/4096 \approx 2.4 \times 10^{-4}$). Cross-validated by 1000-iter sign-permutation in Panel B.

Pre-specification (a priori from thesis, not post-hoc): 6 baselines are the tumor-intrinsic proliferation/repair components defined in §3.3.

Δ SBS5 predicted ↓: radiation kills proliferating cells → their SBS5 mutations clear (higher baseline → more clearance).

Δ IGH clonotypes predicted ↑: radiation-released antigen drives B-cell repertoire expansion (higher baseline → larger antigen pulse → larger Δ IGH).

B

Analytical binomial sign test (one-sided): $P = 0.000244$ ($= 1/4096 = 2.44 \times 10^{-4}$).

Sign-flip permutation null (1000 iterations, seed = 20260527): $P_{\text{empirical}} = < 0.0010$ (0 of 1000 perms reached ≥ 12 concordant; max in null = 11).

Both methods agree to within sampling resolution of 10^{-3} ; the analytical value is the formal P quoted in the manuscript.